

Today

Systems of nonlinear equations

Newton-Raphson for systems

Numerical differentiation

Announcements

- HW 3 due Thurs

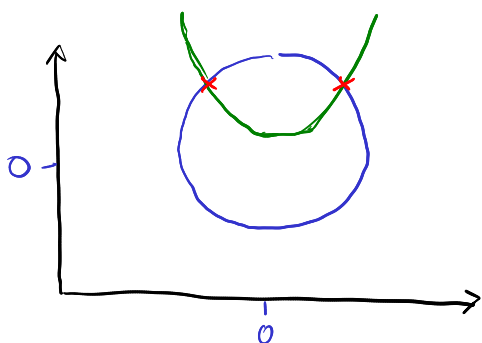
- Exam Next Monday

- Will release Eqn. Sheet/
practice exam today/tomorrow

- Calculator

- Study HW, Labs, Notes

Up to now $f(x) = 0$



$$x_1^2 + x_2^2 = 4$$

$$x_2 = x_1^2$$

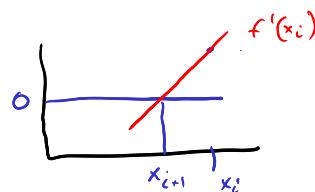
$$\{f\}(\{x\}) = \{0\}$$

$$\vec{f}(\vec{x}) = \vec{0}$$

$$\vec{f}(\vec{x}) = \begin{bmatrix} x_1^2 + x_2^2 - 4 \\ x_2 - x_1^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Newton-Raphson for scalar:

$$f(x_i) = f(x_{i+1}) + f'(x_i)(x_i - x_{i+1})$$



vector case:

first
element
of f $\rightarrow$

choose $\vec{x}_{i+1}$ so this is zero

$$f_1(\vec{x}_i) = f_1(\vec{x}_{i+1}) + \frac{\partial f_1}{\partial x_1}(x_{1,i} - x_{1,i+1}) + \frac{\partial f_1}{\partial x_2}(x_{2,i} - x_{2,i+1}) + \dots + \frac{\partial f_1}{\partial x_n}(x_{n,i} - x_{n,i+1})$$
$$= \sum_{j=1}^n \frac{\partial f_1}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$$f_2(\vec{x}_i) = \sum_{j=1}^n \frac{\partial f_2}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$\vdots$

$$f_n(\vec{x}_i) = \sum_{j=1}^n \frac{\partial f_n}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$$\vec{f}(\vec{x}_i) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \left(\begin{bmatrix} x_{1,i} \\ \vdots \\ x_{n,i} \end{bmatrix} - \begin{bmatrix} x_{1,i+1} \\ \vdots \\ x_{n,i+1} \end{bmatrix} \right)$$

$$\underbrace{\vec{f}(\vec{x}_i)}_{\text{known}} = \underbrace{J}_{\text{known}} (\underbrace{\vec{x}_i}_{\text{known}} - \underbrace{\vec{x}_{i+1}}_{\text{unknown}})$$

$$J^{-1} \vec{f}(\vec{x}_i) = \vec{x}_i - \vec{x}_{i+1}$$

$$\boxed{\vec{x}_{i+1} = \vec{x}_i - J^{-1} \vec{f}(\vec{x}_i)}$$

(scalar)

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

want $\vec{x}_{i+1} = \vec{x}_i + \Delta \vec{x}$
 $\Delta \vec{x} = \vec{x}_{i+1} - \vec{x}_i$

solve $J \Delta \vec{x} = -\vec{f}(\vec{x}_i)$
 $\vec{x}_{i+1} \leftarrow \vec{x}_i + \Delta \vec{x}$

or

$\vec{x}_{i+1} \leftarrow \vec{x}_i - J \backslash \vec{f}(\vec{x}_i)$

$A \backslash b \equiv \text{solution to } Ax = b$

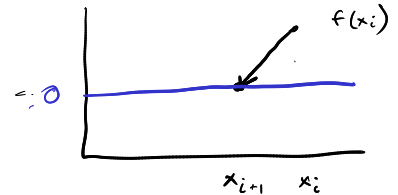
Newton-Raphson for Systems

input : function $\{f\} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and its Jacobian $[J]$ (with $J_{ij} = \partial f_i / \partial x_j$), initial guess $\{x\}_0$, tolerance ε_s , maximum iterations N

output: estimate $\{x\}_r$ of the root

```
{x}_r ← {x}_0
for i ← 1 to N do
    {x}_{r,old} ← {x}_r
    solve [J]({x}_{r,old}) {Δx} = -{f}({x}_{r,old})
    {x}_r ← {x}_{r,old} + {Δx}
    ε_a ← (||{x}_r - {x}_{r,old}|| / ||{x}_r||) · 100%
    if ε_a < ε_s then
        return {x}_r
    end
end
return {x}_r // did not converge in N iterations
```

(scalar)



Two Challenges

1. Convergence

2. Calculating Jacobian

Optimization

minimize $\sum_{k=1}^n (f_k(\vec{x}))^2$

Strength

"quadratic convergence" near root

Calculating Derivatives

Symbolic

- By hand
- Computer Algebra System (CAS)
 - e.g. Symbolic Toolbox
- Automatic Code gen

Numerical

Automatic

- Tracks exact derivative through each operation

Numerical Differentiation

Forward: $f'(x) \approx \frac{f(x+h) - f(x)}{h}$

Central: $f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$ ← more accurate

Column of Jacobian → $\frac{\partial \vec{f}}{\partial x_j} \approx \frac{\vec{f}(\vec{x} + h\vec{e}_j) - \vec{f}(\vec{x})}{h}$

size of h ?

Numerical Jacobian (forward difference)

input : function $\{f\} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, point $\{x\}$, step size h

output: approximation $[J]$ of the Jacobian at $\{x\}$, $J_{ij} \approx \partial f_i / \partial x_j$

$\{f\}_0 \leftarrow \{f(\{x\})\}$

for $j \leftarrow 1$ to n do

$\{e_j\} \leftarrow$ unit vector in the j direction // j th column of $[I]$

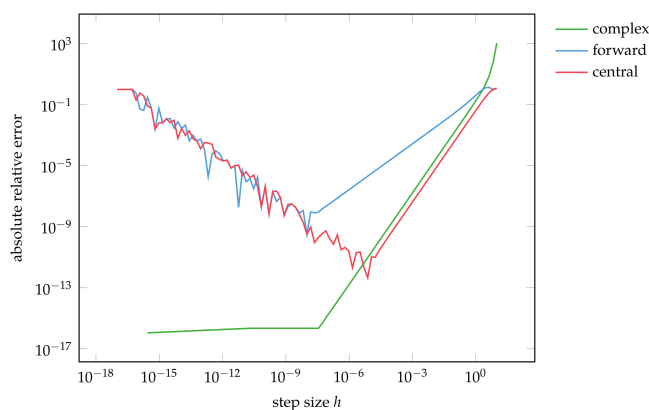
$[J][:, j] \leftarrow \frac{\{f(\{x\} + h\{e_j\})\} - \{f\}_0}{h}$ // j th column of $[J]$

end

return $[J]$

$$h \approx \sqrt{\epsilon} \times$$

↑
machine precision ϵ



Taylor Series

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2} + \dots$$

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \underbrace{\frac{f''(x)h}{2} + \dots}_{\text{Truncation Error}}$$

Truncation Error

Roundoff Error

$$\frac{2\epsilon |f(x)|}{h}$$

Best point is where these are equal

$$\frac{2\epsilon |f(x)|}{h} = \frac{f''(x)h}{2}$$

$$\frac{4|f(x)|}{f''(x)} \epsilon = h^2$$

$$h \approx \sqrt{\epsilon} \times$$

(right order of magnitude)